首都师范大学
CAPITAL NORMAL UNIVERSITY

电路分析

线性电路定理

主讲人：朱 虹

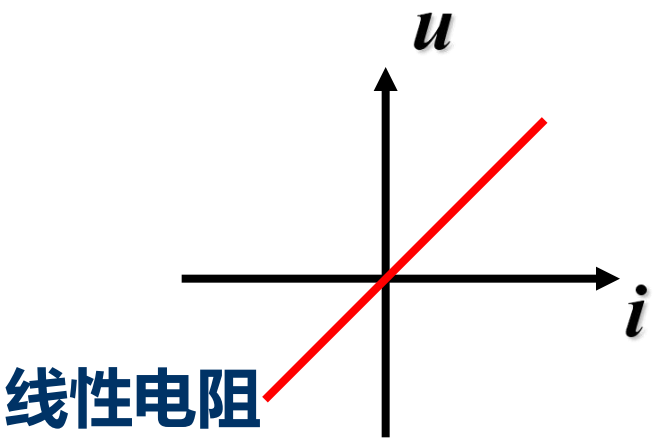
一 → 线性电路（linear circuits ）

线性元件 VAR 是直线

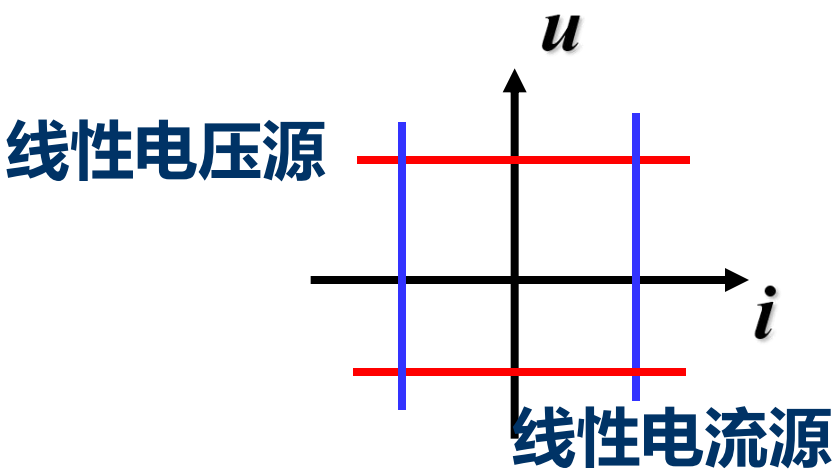
$u = Ri$ ***R*** **是定值**

$u = \alpha i$ ***α*** **是定值**

线性受控源



线性电源 VAR 是直线



线性电路（Linear circuits）

线性元件 + 线性电源

线性电路定理（Theorems for linear circuits）

叠加定理（Superposition theorem ）

齐次定理（Homogeneity theorem ）

戴维南定理（Thevenin theorem ）

诺顿定理（Norton theorem ）

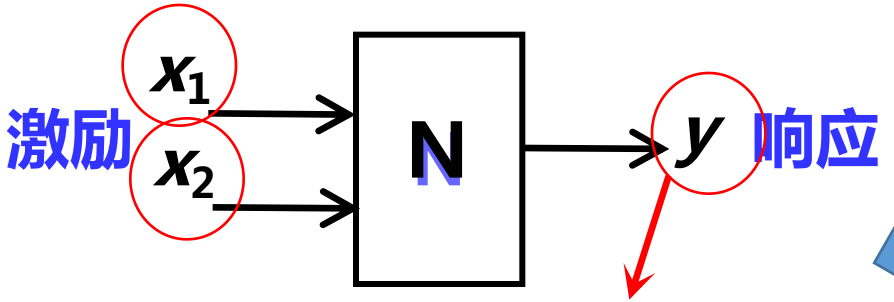
最大功率传输定理（Maximum power transfer ）

二 → 叠加定理（Superposition theorem ）

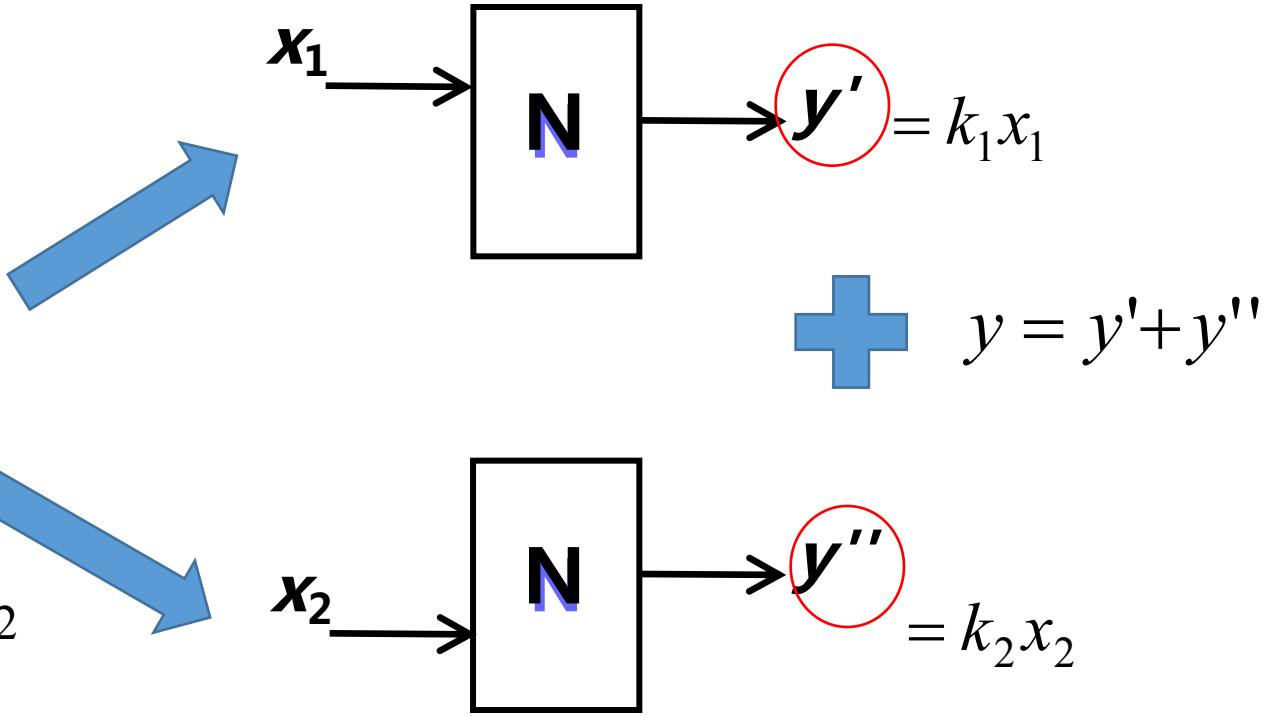
1 叠加定理

线性电路 $y = kx_1$ **齐次性**

$y = k_1x_1 + k_2x_2$ **可加性**



$y = k_1x_1 + k_2x_2$



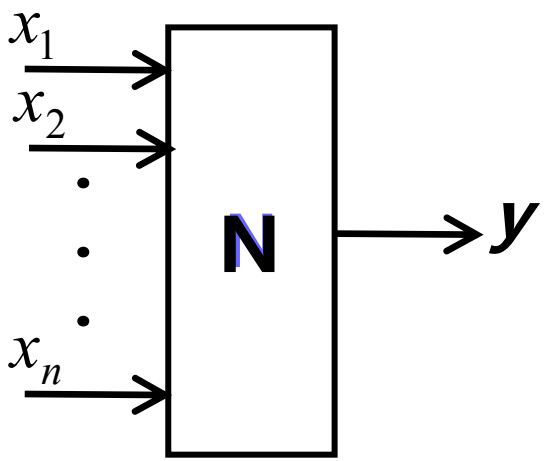
对于线性电路，两个激励共同作用的响应，

等于各激励单独作用时的响应的代数和

推广：

n **个激励** $x_1, x_2, \cdots x_n$

每个激励单独作用的响应 $y_1, y_2, \cdots y_n$



$$y_1 = k_1 x_1 \qquad y_2 = k_2 x_2 \qquad \cdots \qquad y_n = k_n x_n$$

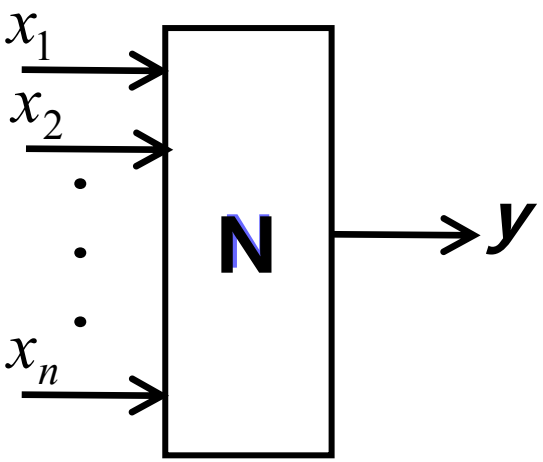
n **个激励共同作用的响应**

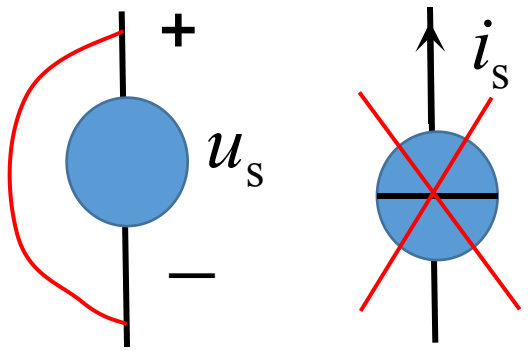
$$y = k_1 x_1 + k_2 x_2 + \cdots + k_n x_n \quad = y_1 + y_2 + \cdots y_n = \sum_{i=1}^{n} y_i$$

对于线性电路，*n*个激励共同作用的响应，

等于各激励单独作用时的响应的代数和

$$y=\sum_{i=1}^{n} y_i=\sum_{i=1}^{n} k_i x_i$$





x_i **激励：** **电路中的独立源**

电压源、电流源

y_i, y **响应：** **支路电流、电压**

k_i **系数** **由线性电路N决定**

注意： **1. 功率不能是响应** $p=ui$ **功率不是激励的线性函数**

2. 受控源不能是激励 **把受控源当作元件**

3. 某激励单独作用，其他激励取0

二 → 叠加定理（Superposition theorem ）

2 叠加定理例题1

例1 用叠加定理计算 3Ω 电阻的电压及功率。

解：1. 电压源单独激励

$$U'=\frac{3}{6+3}\times(-12)=-4\,(\mathrm{V}) \qquad P'=\frac{U'^2}{3}=\frac{16}{3}\,(\mathrm{W})$$

2. 电流源单独激励

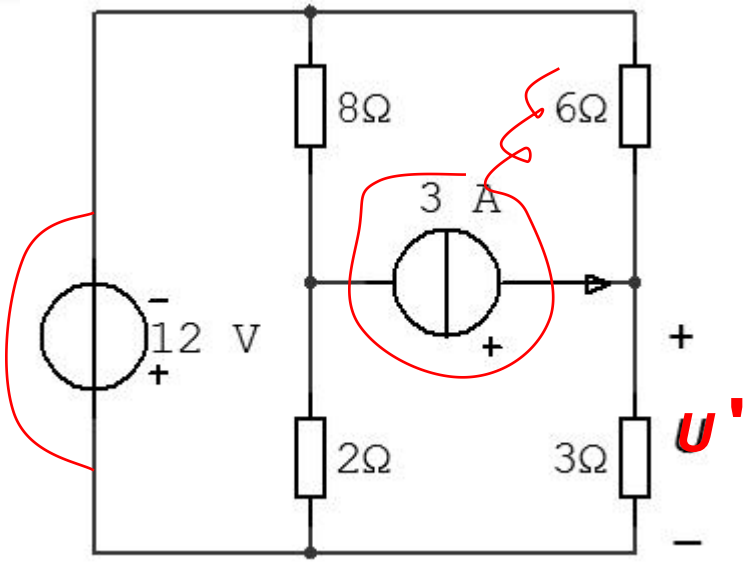
$$I_1=\frac{6}{3+6}\times 3=2\,(\mathrm{A}) \qquad P''=\frac{U''^2}{3}=12\,(\mathrm{W})$$

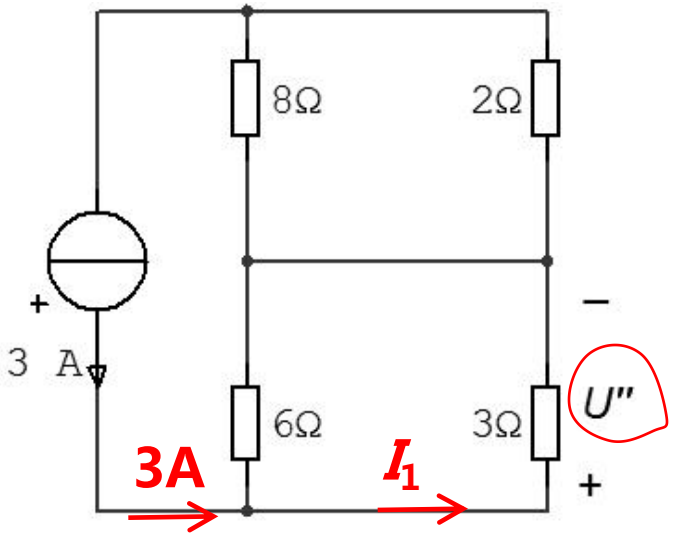
$$U''=3I_1=3\times 2=6\,(\mathrm{V})$$

3. 叠加

$$U=U'+U''=-4+6=2\,(\mathrm{V})$$

$$P=\frac{U^2}{3}=\frac{4}{3}\,(\mathrm{W}) \qquad P=P'+P''=17\frac{1}{3}\,(\mathrm{W})$$





二 → 叠加定理（Superposition theorem ）

3 叠加定理例题2

例 2 用叠加定理计算受控源的电压及功率

解：1. 电压源单独激励

$$(1+2)I'+3(I'+2I')=4 \qquad I'=\frac{1}{3}\,\mathrm{A}$$

$$U'=3\cdot 3I'=9I'=3\,(\mathrm{V})$$

2. 电流源单独激励

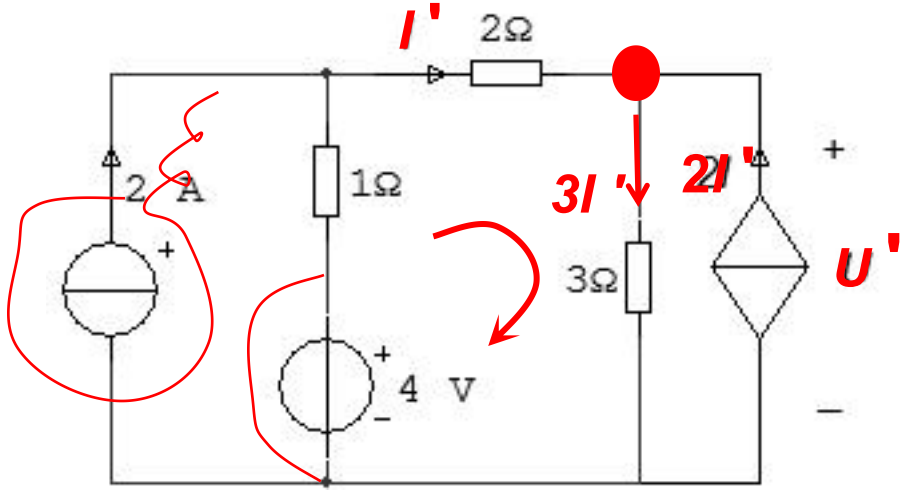
$$2I''+3\cdot 3I''=1\,(2-I'')$$

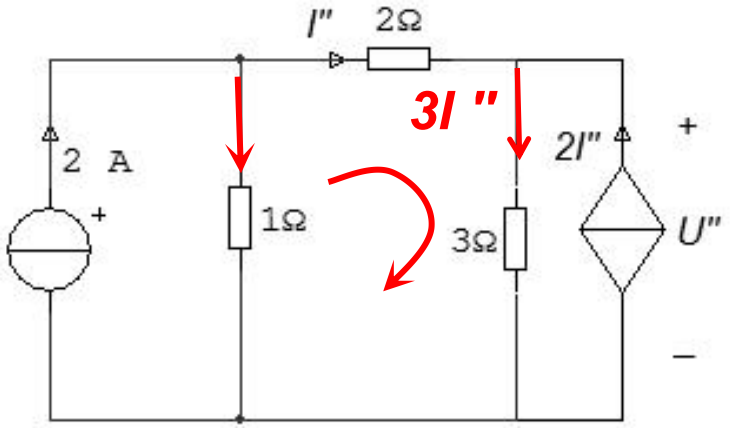
$$I''=\frac{1}{6}\,\mathrm{A} \qquad U''=9I''=1.5\,(\mathrm{V})$$

$$U=U'+U''=4.5\,(\mathrm{V})$$

$$I=I'+I''=0.5\,(\mathrm{A})$$

$$P=-U\cdot(2I)=-4.5\,(\mathrm{W})$$

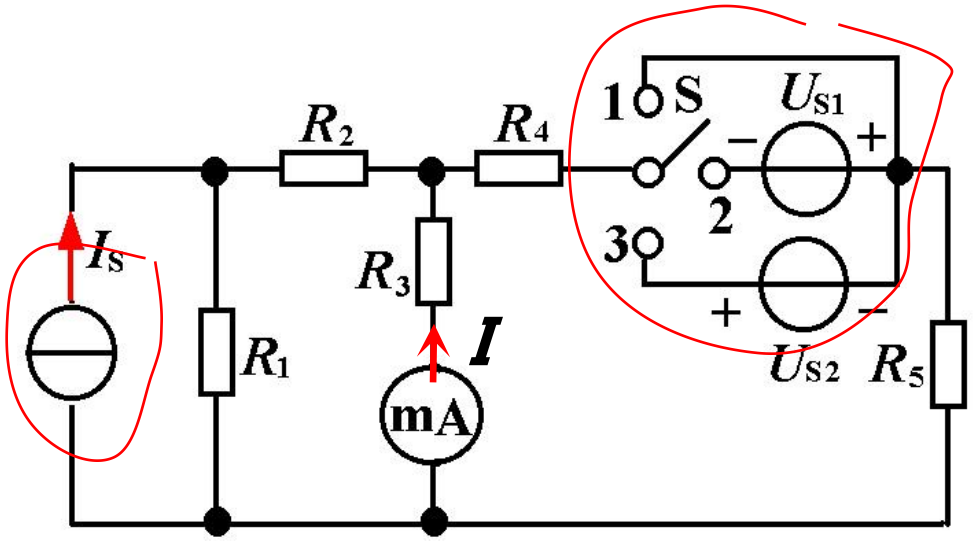




二 → 叠加定理（Superposition theorem ）

4 叠加定理例题3

例3　U_{s1} = 10V，U_{s2} = 15V，当S在位置1时，毫安表的读数为I' = 40mA；当开关S合向位置2时，毫安表的读数为I'' = - 60mA。如果把开关S合向位置3，则毫安表的读数 I 为多少？



S=1　I_s 单独激励

$$I' = k_1 I_s = 40$$

S=2　I_s , U_{s1} 共同激励

$$I'' = k_1 I_s + k_2 U_{s1} = -60$$

S=3　I_s , U_{s2} 共同激励　$U_{s2} = -15\,\mathrm{V}$

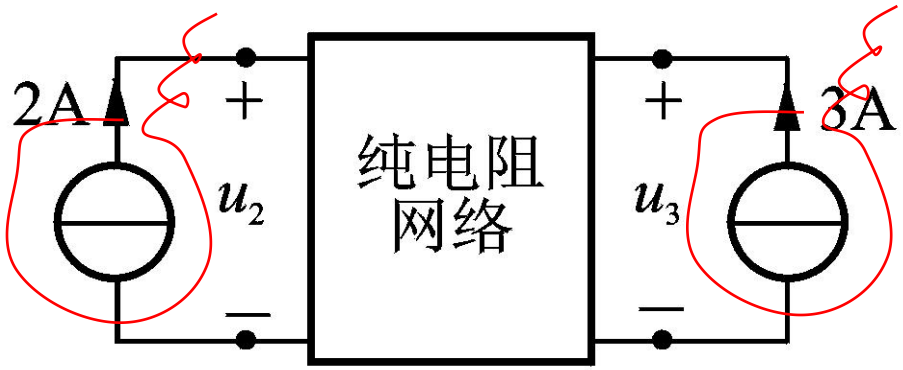
$$\begin{cases} k_1 I_s = 40 \\ k_1 I_s + 10 k_2 = -60 \end{cases} \qquad k_2 = -10$$

$$I = k_1 I_s + k_2 U_{s2} = 40 + (-10) \times (-15) = 190\ (\mathrm{mA})$$

二 → 叠加定理（Superposition theorem ）

5 叠加定理例题4

例4 当2A电流源未接入时，3A电流源向网络提供的功率为54 W，u_2 = 12 V；当3A未接入时，2A电流源向网络提供的功率为28 W，u_3 = 8 V。求两电源同时接入时，各电流源的功率。



解：1. 3A单独激励

$$U_2' = 12\,\text{V}$$

$$3U_3' = 54 \qquad U_3' = 18\,\text{V}$$

2. 2A单独激励

$$U_3'' = 8\,\text{V}$$

$$2U_2'' = 28 \qquad U_2'' = 14\,\text{V}$$

3. 共同激励

$$U_2 = U_2' + U_2'' = 12 + 14 = 26\,(\text{V})$$

$$U_3 = U_3' + U_3'' = 18 + 8 = 26\,(\text{V})$$

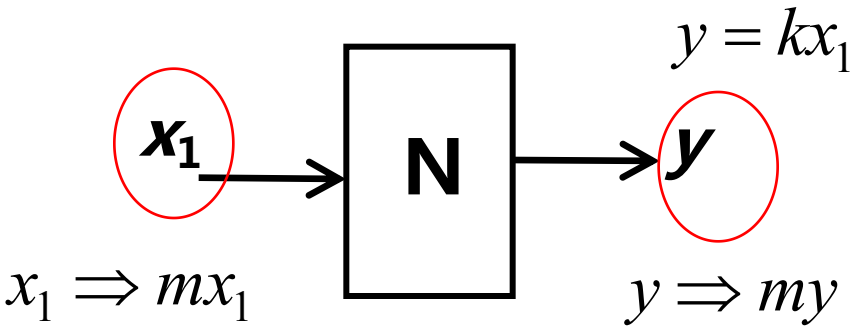
$$P_{2\text{A}} = -2U_2 = -52\,(\text{W})$$

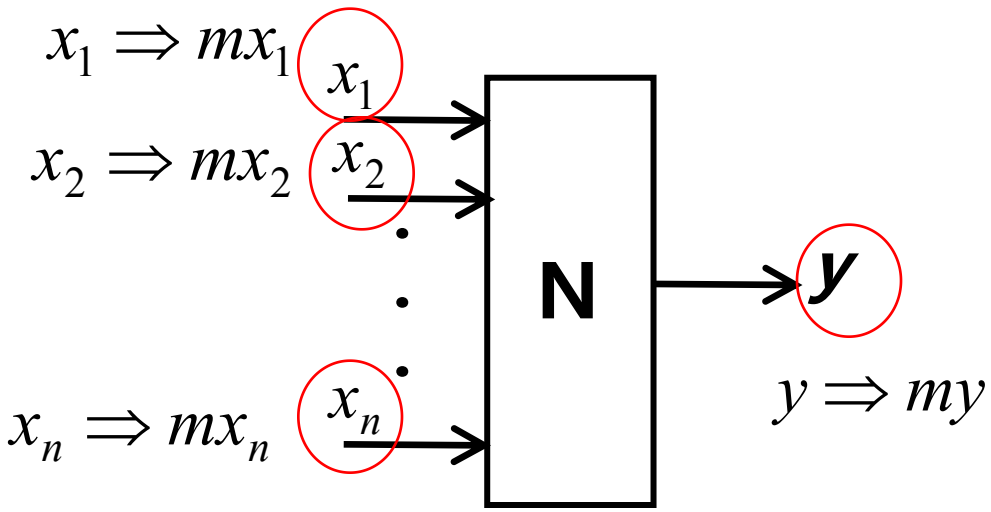
$$P_{3\text{A}} = -3U_3 = -78\,(\text{W})$$

三 → 齐次定理（Homogeneity theorem ）

1 齐次定理

线性电路　齐次性





对于线性电路，每个激励同时增加/缩小 m 倍，响应也增加/缩小 m 倍

三 → 齐次定理（Homogeneity theorem ）

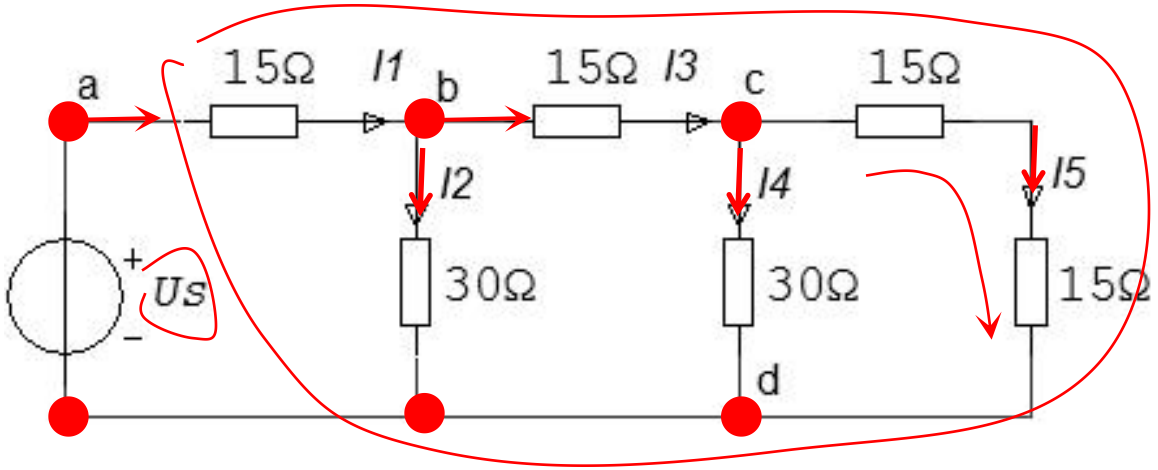
2 齐次定理例题

例1 已知U_s=6V, 用齐次定理计算电流 I_5

$$I_5 = kU_s = 6k$$

$$k = \frac{I_5}{U_s} = \frac{I_5'}{U_s'}$$

$$I_5' \Rightarrow U_s'$$



倒推 $I_5' = 1\,\text{A}$ $\quad U_{cd} = (15+15)I_5' = 30\,(\text{V})$ $\quad I_4' = \dfrac{U_{cd}}{30} = 1\,(\text{A})$

$I_3' = I_4' + I_5' = 2\,(\text{A})$ $\quad U_{bd} = 15I_3' + 30I_4' = 60\,(\text{V})$

$I_2' = \dfrac{U_{bd}}{30} = 2\,(\text{A})$ $\quad I_1' = I_2' + I_3' = 4\,(\text{A})$ $\quad U_s' = 15I_1' + 30I_2' = 120\,(\text{V})$

$$k = \frac{I_5'}{U_s'} = \frac{1}{120} \qquad I_5 = 6k = \frac{6}{120} = 0.05\,(\text{A})$$

四 → 戴维南定理（Thevenin theorem ）

1 戴维南定理

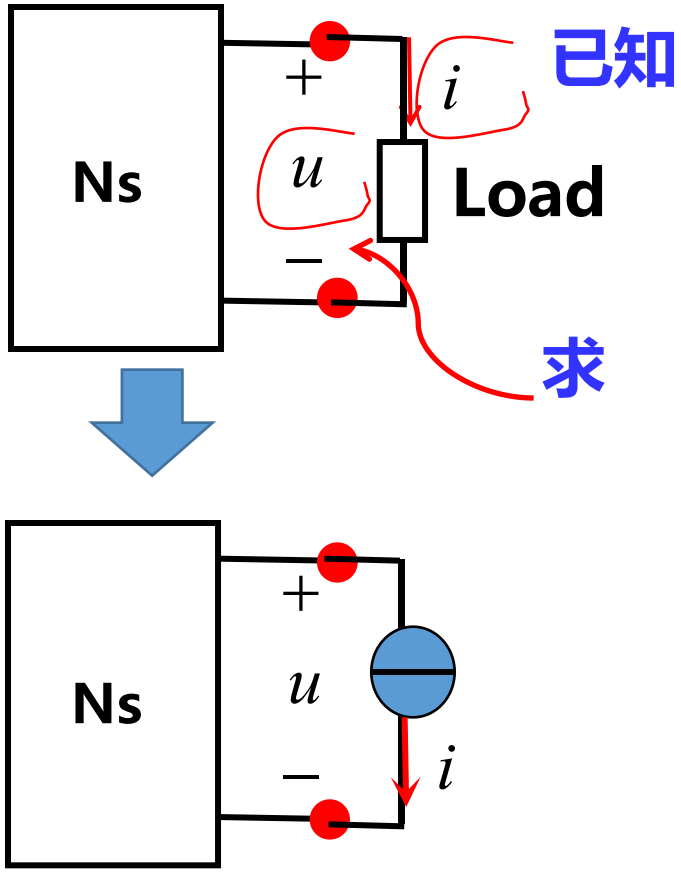
线性含源二端网络Ns

线性电阻

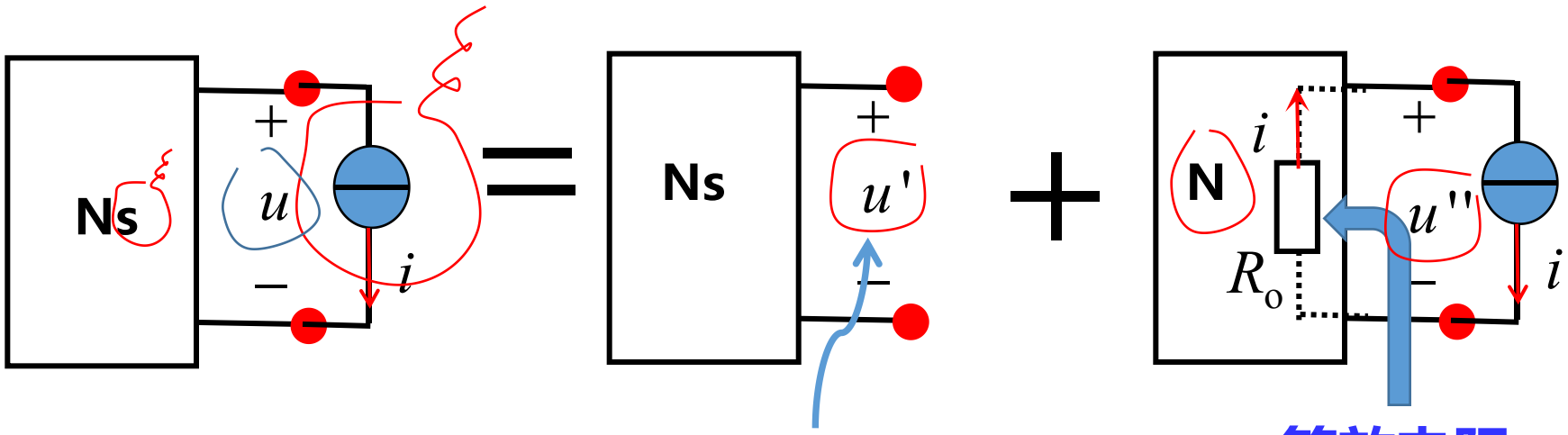
线性受控源

线性电压源

线性电流源



$$u = \mathrm{f}\,(i)$$



$$u = u' + u'' = u_{oc} - R_o i$$

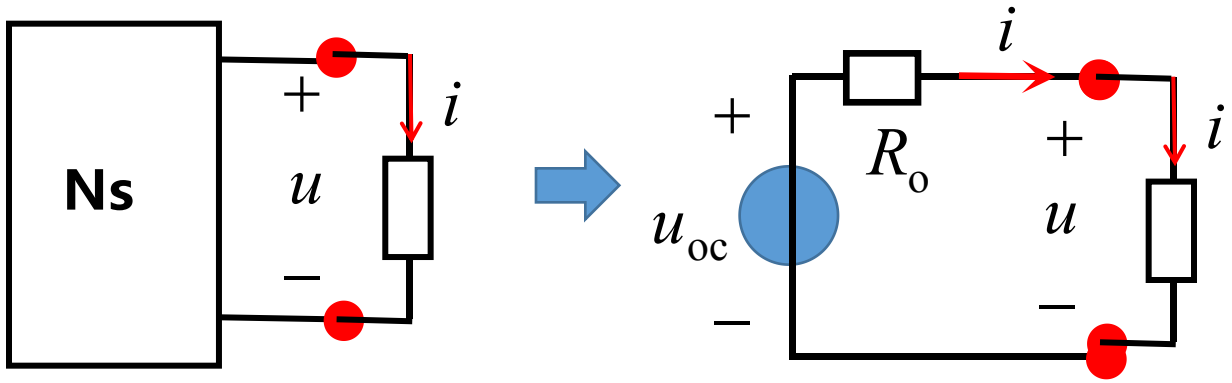
开路电压 u_{oc}

$$u' = u_{oc}$$

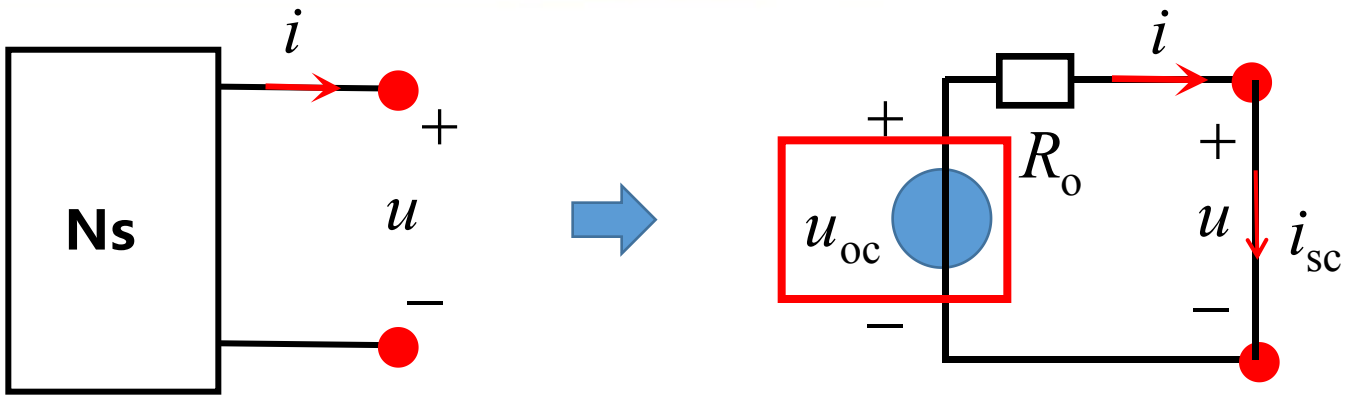
等效电阻

输出电阻

$$u'' = -R_o i$$



戴维南定理

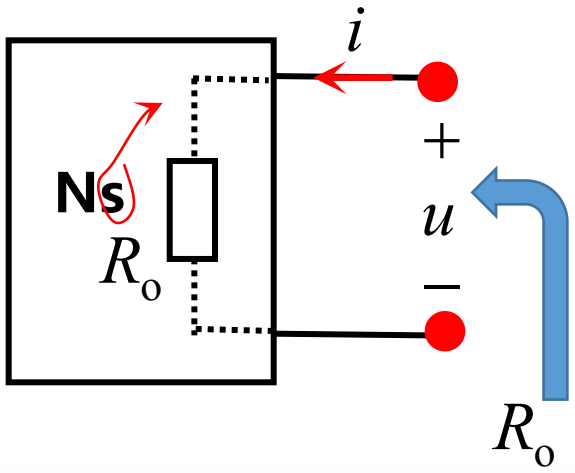


电压源的电压是该二端网络的开路电压。u_{oc}

等效电阻是该二端网络独立源取0的等效电阻。R_o

等效电阻的求法：

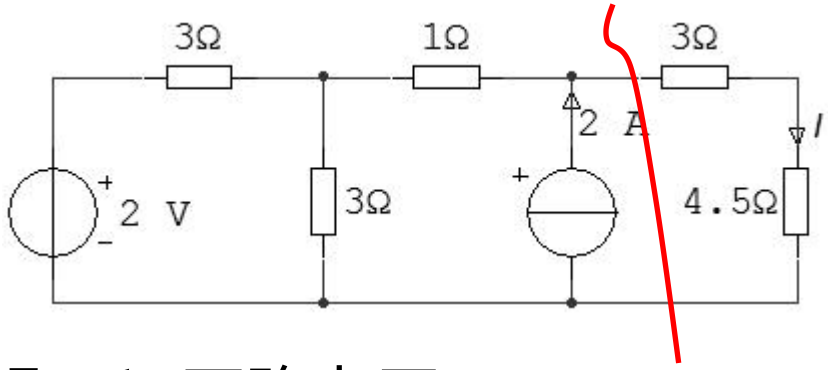
1. 电阻串并联
2. 外加电压法 $R_o = \dfrac{u}{i}$
3. 短路电流法 $R_o = \dfrac{u_{oc}}{i_{sc}}$

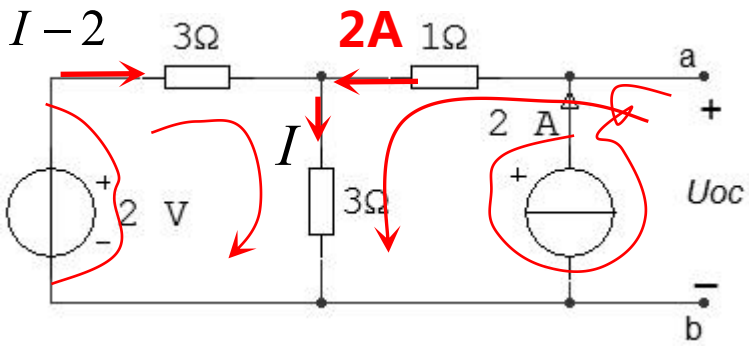


四 → 戴维南定理（Thevenin theorem ）

2 戴维南定理例题1

例1 用戴维南定理计算电流 I。



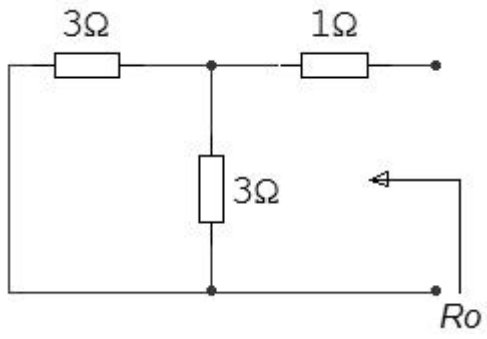


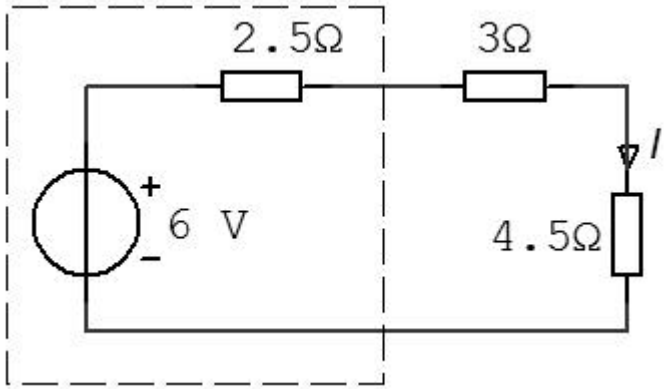
解：1 开路电压 U_{oc}

$$3(I-2)+3I=2 \qquad I=\frac{4}{3}\,\text{A}$$

$$U_{oc}=1\times 2+3I=6\,(\text{V})$$

2 等效电阻 R_o





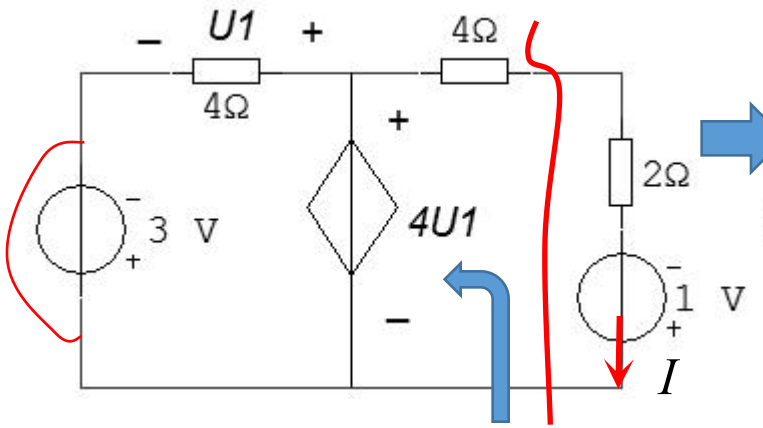
$$R_o=1+3//3=2.5\,(\Omega)$$

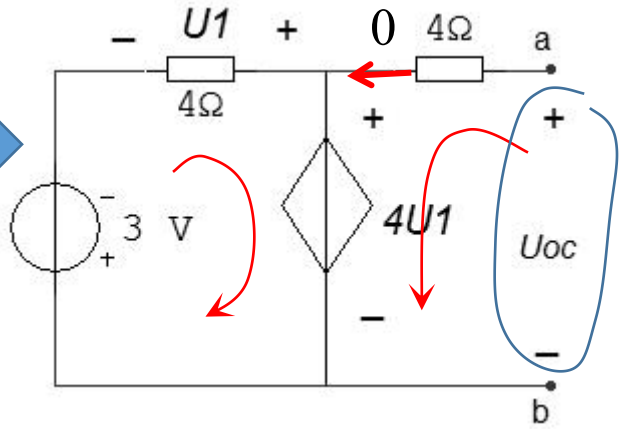
$$I=\frac{6}{2.5+3+4.5}=0.6\,(\text{A})$$

四 → 戴维南定理（Thevenin theorem ）

3 戴维南定理例题2

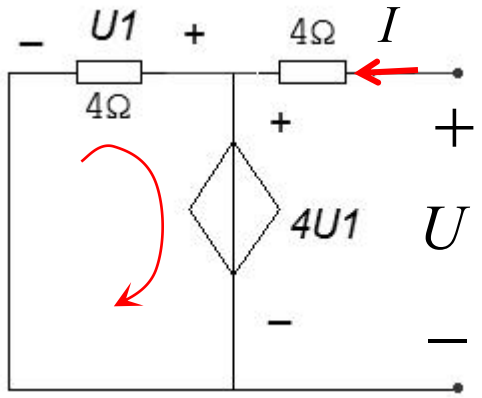
例2 用戴维南定理求1V电压源的功率。





$$-U_1+4U_1+3=0 \quad U_1=-1\,\text{V}$$

$$U_{oc}=4U_1=-4\,(\text{V})$$

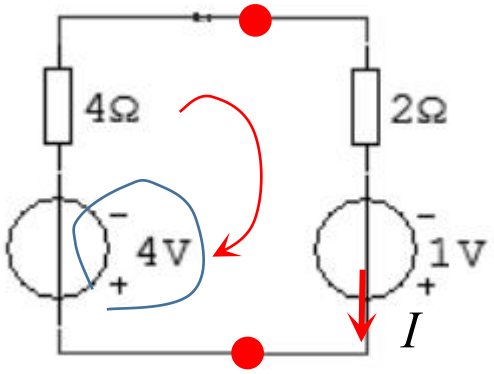


外加电压法

$$U_1=4U_1 \qquad U_1=0$$

$$U=4I+4U_1 \quad =4I$$

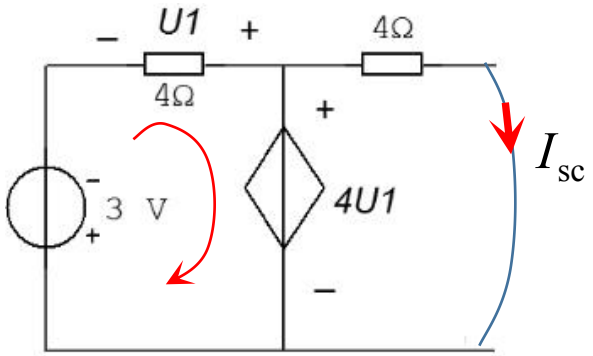
$$R_o=\frac{U}{I}=4\,(\Omega)$$



$$(4+2)I-1+4=0 \qquad I=-0.5\,\text{A}$$

$$P=-1\cdot I=0.5\,(\text{W})$$

短路电流法



$$-U_1+4U_1+3=0$$

$$U_1=-1\,\text{V}$$

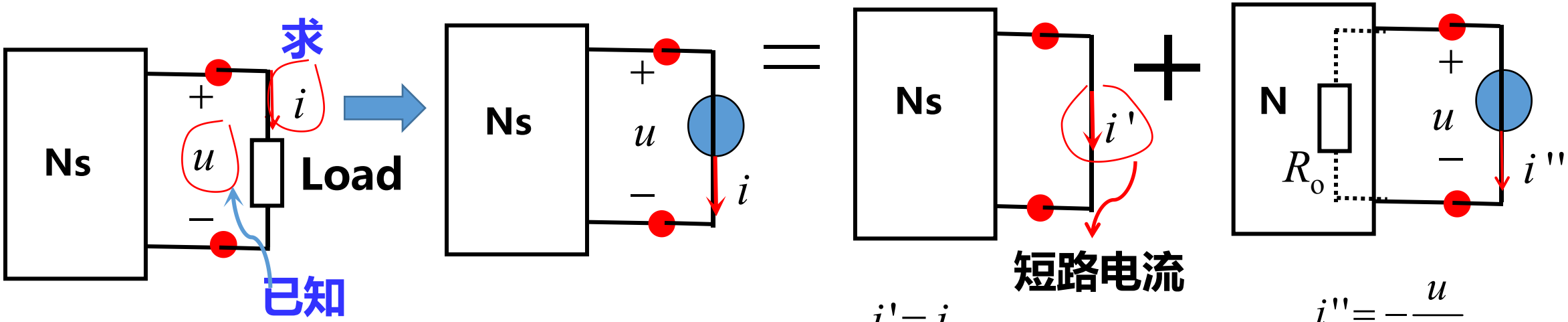
$$I_{sc}=\frac{4U_1}{4}=-1\,(\text{A})$$

$$R_o=\frac{U_{oc}}{I_{sc}}=\frac{-4}{-1}=4\,(\Omega)$$

五 → 诺顿定理（Norton theorem ）

1 诺顿定理

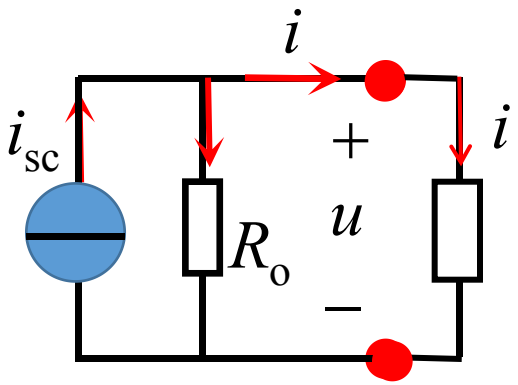
线性含源二端网络Ns　　$u = \mathrm{f}\,(i)$



$$i = i' + i'' = i_{sc} - \frac{u}{R_o}$$

$$i' = i_{sc}$$

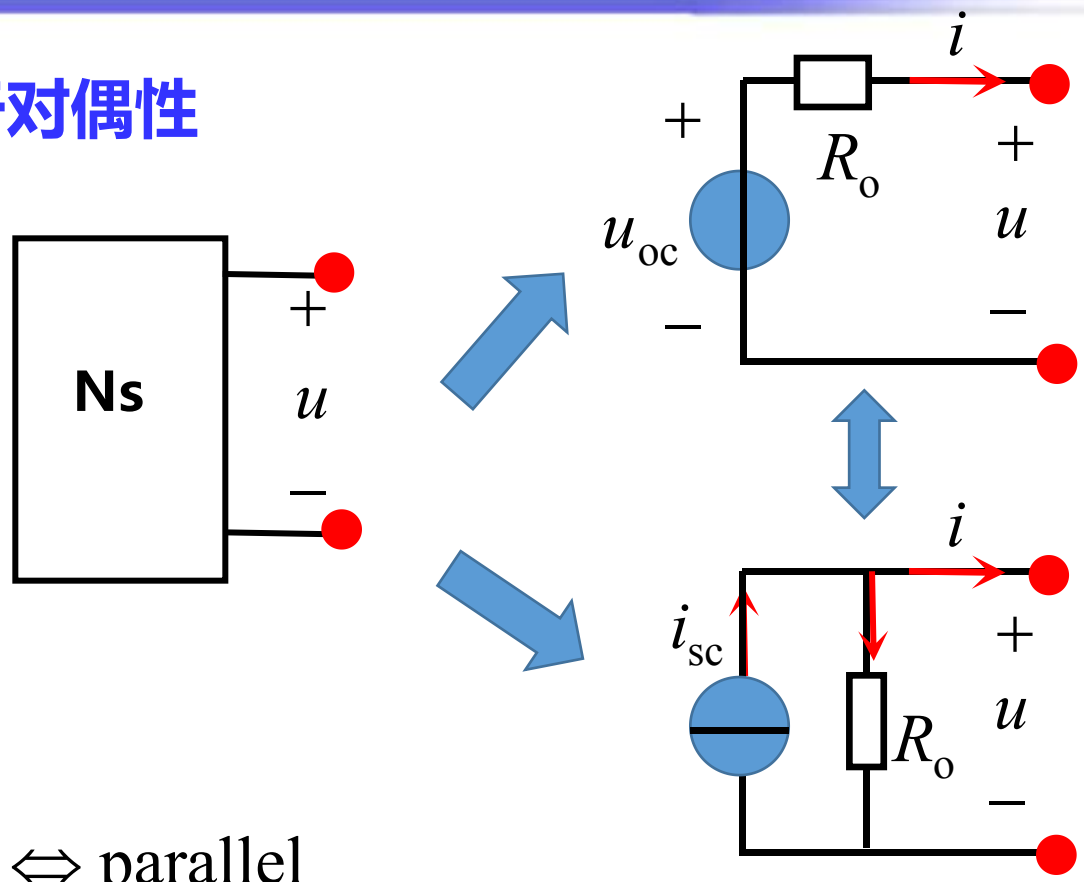
$$i'' = -\frac{u}{R_o}$$



五 → 诺顿定理（Norton theorem ）

2 对偶性

关于对偶性



$$u = u_{oc} - R_o i$$

$$i = i_{sc} - G_o u$$

serial $\Leftrightarrow$ parallel

$u \Leftrightarrow i$ oc $\Leftrightarrow$ sc $u_{oc} \Leftrightarrow i_{sc}$

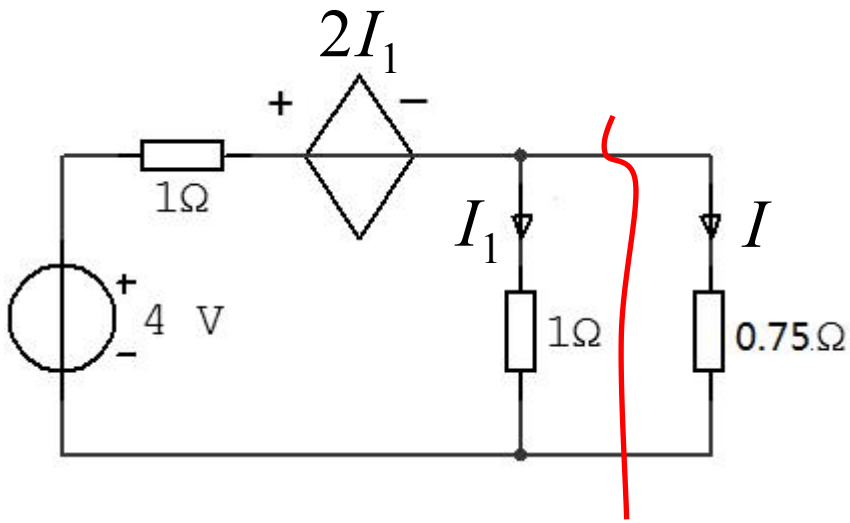
$R \Leftrightarrow G$

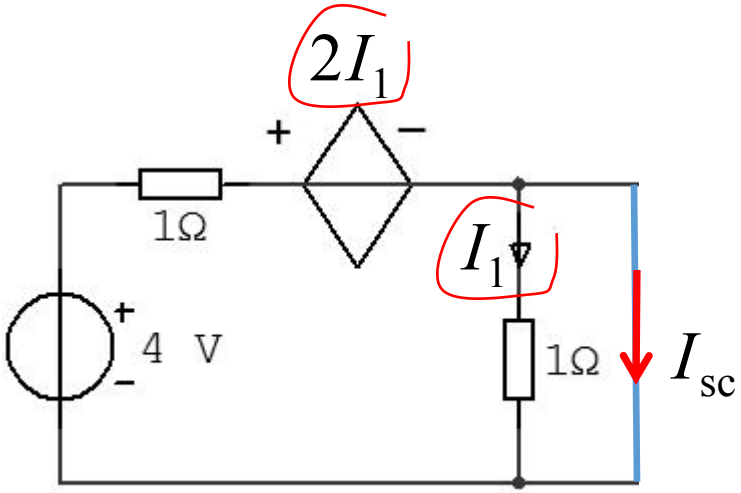
Thevenin $\Leftrightarrow$ Norton

五 → 诺顿定理（Norton theorem ）

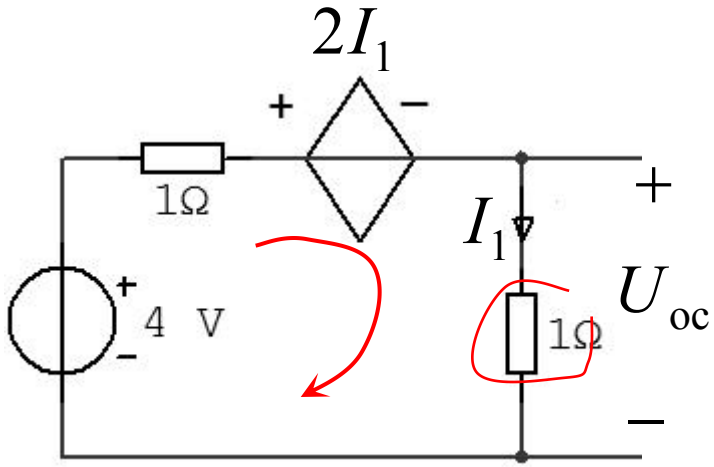
3 诺顿定理例题

例1　用诺顿定理计算电流 I。





$$I_{sc}=\frac{4}{1}=4\,(\mathrm{A})$$

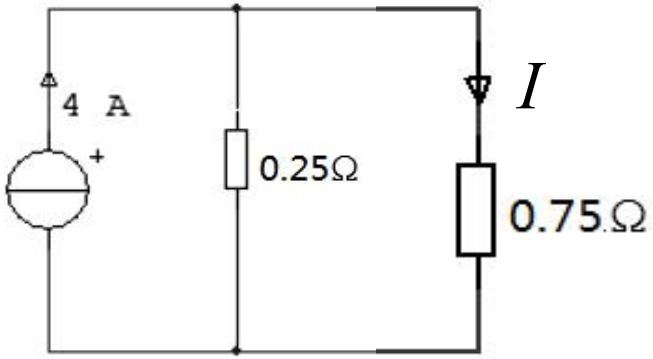


$$(1+1)I_1+2I_1=4$$

$$I_1=1\,\mathrm{A}$$

$$U_{oc}=1\cdot I_1=1\,(\mathrm{V})$$

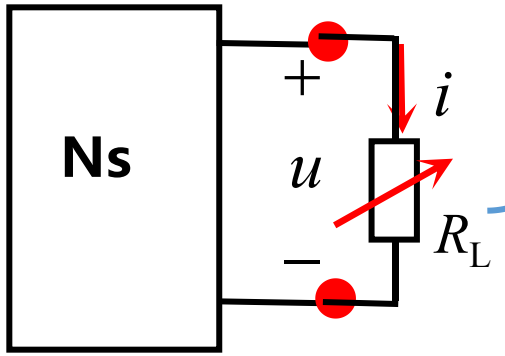
$$R_o=\frac{U_{oc}}{I_{sc}}=\frac{1}{4}=0.25\,(\Omega)$$



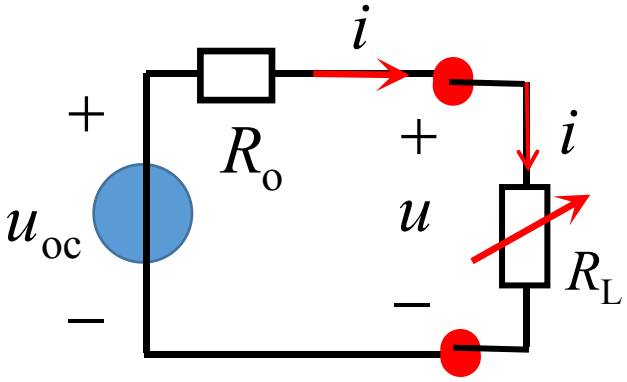
$$I=\frac{0.25}{0.25+0.75}\times 4=1\,(\mathrm{A})$$

六 → 最大功率传输定理
（Maximum power transfer ）

1 最大功率传输定理



$p_L \to \max$



$$R_L = R_o \quad p_L \to \max$$

$$p_{Lmax} = \frac{u_{oc}^2}{4R_o}$$

$$i = \frac{u_{oc}}{R_o + R_L} \qquad p_L = R_L i^2 = u_{oc}^2 \cdot \frac{R_L}{(R_o + R_L)^2}$$

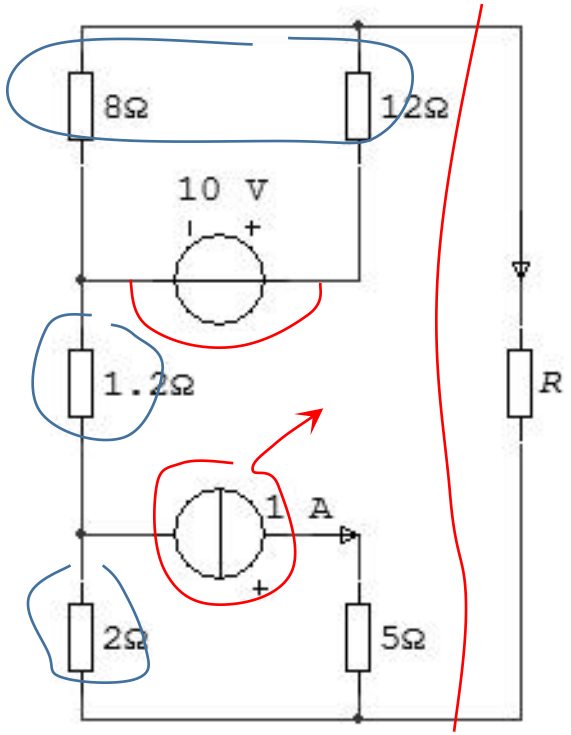
$$\frac{dp_L}{dR_L} = u_{oc}^2 \cdot \frac{(R_o + R_L)^2 - R_L \cdot 2(R_o + R_L)}{(R_o + R_L)^4} = u_{oc}^2 \cdot \frac{R_o^2 - R_L^2}{(R_o + R_L)^4} = u_{oc}^2 \cdot \frac{R_o - R_L}{(R_o + R_L)^3} = 0$$

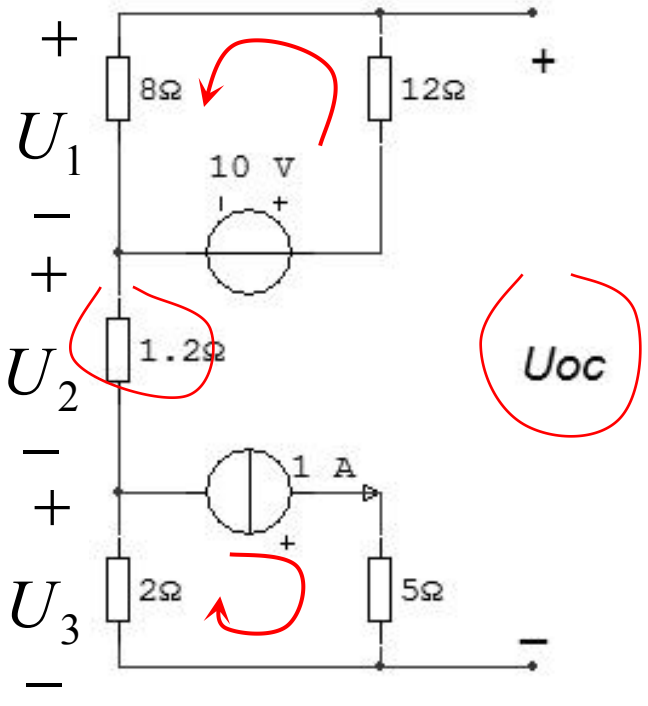
$$p_o = p_L = \frac{u_{oc}^2}{4R_o} \qquad \eta = 50\%$$

六 → 最大功率传输定理
（Maximum power transfer ）

2 最大功率传输例题

例1 求电阻R获得最大功率时的电阻，并求最大功率。



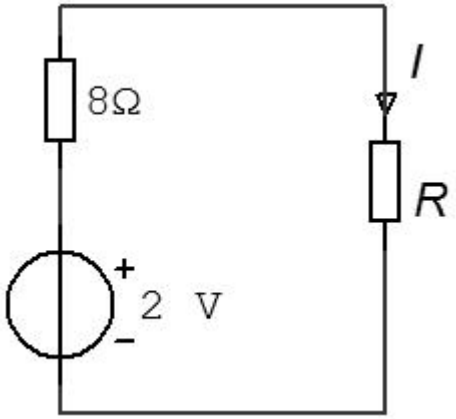


$$U_1 = \frac{8}{8+12} \times 10 = 4\,(\mathrm{V})$$

$$U_2 = 0$$

$$U_3 = -1 \times 2 = -2\,(\mathrm{V})$$

$$U_{oc} = U_1 + U_2 + U_3 = 2\,(\mathrm{V})$$



$$R_o = 8//12 + 1.2 + 2 = 8\,(\Omega)$$

$$R = R_o = 8\,\Omega$$

$$P_{max} = \frac{u_{oc}^{\ 2}}{4R_o} = \frac{2^2}{4 \times 8} = 0.125\,(\mathrm{W})$$

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